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Studies on biosynthesis of xanthan gum using *Xanthomonas sp.*, isolated from infected cotton leaves

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Abstract Xanthan is a vital biopolymer synthesized by *Xanthomonas campestris* and possesses loads of business significance. Naturally occurring strains of *Xanthomonas* like *X.campestris* was isolated from the cotton leaves tainted with angular leaf spot and was recognized by biochemical characterization. Xanthan gum is produced by the utilizing distinctive carbon sources like Sucrose, Maltose, Lactose and Molasses. The production of biomass was carried on by incubating at 27°C of pH 6.0 and Xanthan gum synthesis was carried on at 30°C of pH 7.0, respectively. The produced gum was precipitated by the addition of ethanol (3:1 v/v) and dried in hot air oven at 40°C. The dry weight of biomass was found by drying the pellet in hot air oven at 105°. The structural and functional characterization of the Xanthan gum was dictated by SEM examination and FTIR respectively. The present work focused on the utilization of various carbon sources on the production of xanthan gum by utilizing *Xanthomonas* strains.

Key Words Xanthan gum, Fermentation, *Xanthomonas sp.*, Molasses, SEM, FTIR

Introduction

Xanthan gum is an extracellular polysaccharide produced by a few types of *Xanthomonas* like *Xanthomonas campestris*, *X.phaseoli* and *X.malvacearum*. Eventhough they are created by every one of these *Xanthomonas sp.*, the gum delivered by *X. campestris* is having sure rheological properties of business centrality. On account of its exceptional rheological properties, xanthan is utilized as a part of nourishment, pharmaceuticals, beautifiers, paper, paint, materials, cements and furthermore in oil and gas industry (Sutherland., 1996; Kennedy and Bradshaw., 1984). The security and toxicological properties of xanthan gum for their applications in sustenance and pharmaceutical industry have been widely considered. The gum possess huge applications in variety of fields as a viscosifying agent and suspending operator (Cottrell and Kang et al., 1975; Margaritis and Pace 1985)

The second microbial polysaccharide popularized was xanthan gum which is an imperative mechanical biopolymer. Some natural limits, attributed to the exopolysaccharide produced by these pathogenic microorganisms, contain protection against environmental conditions, for instance, drying, temperature changes, radiation, certain chemical compound and adhesion (Romeiro., 1995).

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Xanthan structure depends on cellulose spine having exchange glucosyl buildups substituted by a triasaccharide chain made out of D-mannose, D-glucouronic acid and a terminal D-mannose. . It is a commercial polysaccharide and an industrially important biopolymer that is confirmed to be an attractive alternative for taking the place of chemically extracted traditional gums acquired from plants and marine algae. Xanthan possess remarkable properties like high viscosity even at low concentrations, pseudoplasticity, and insensitivity to a wide range of temperature, pH, and electrolyte concentrations. Xanthan is toxic free and non sensitizing, so does not bring about skin or eye disturbance. In light of this property, xanthan has been affirmed by the Food and Drug administration,USA to use as a nourishment added substance with no particular amount impediments (Kennedy and Bradshaw1984; Rosalam and England 2006).

Overall utilization of xanthan is around 23 million kg/year and is assessed to develop at a yearly rate of 5-10% ceaselessly (Moreira et al., 2001). Synthesis of xanthan is affected by the utilization of different production media components (Garcia-Ochoa et al., 2001; Aarthy Palaniraj et al,2011; Amanullah et al,1998). Most methods of commercial xanthan gum production employs glucose or invert sugars, and many industries prefer batch processes than the continuous fermentation processes.(Letisse et al,2001). Instead of glucose or sucrose, the use of molasses may results in the cheaper cost of the final product. Agri-nourishment by-items rich in sugars can likewise be utilized to deliver high esteem included sustenance fixings, for example, xanthan gum. In this current situation, byproducts of sugarcane are the enthralling substrate for production of xanthan gum. Sugar beet molasses is most widely utilized in the fermentation process since it can also behave as growth factors like pantothenic acid, inositol and so on. (Antunes AEC et al.2003).

The present investigation focused for the examination of different carbon sources in xanthan gum production by using *X. campestris* isolates. Thus, in this work the product of the wild strains isolated from the infected cotton plant was screened for generation of xanthan gum by using different carbon sources.

Materials and methods

Isolation of *Xanthomonas sp.*, from infected cotton leaves

Xanthomonas sp., is isolated from the leaf segment of the cotton plant infected by bacterial angular leaf spot of cotton (*Gossypium hirsutum*). A small portion of the infected cotton leaves were excised and surface sterilized with 0.1% $HgCl_2$.The extracted leaves were placed on the autoclaved YM agar plates containing yeast extract, 3 (g L^{-1}); malt extract, 3 (g L^{-1}) ; peptone, 5 (g $L^{-1)}$; glucose, 10(g L^{-1}); agar, 20 (g L^{-1}) and were incubated at 28°C for 24 hrs. (Jeanes et.al., 1976). The mucoid colonies obtained were sustained on yeast agar slants and stored at 8°C.

Morphological and biochemical characteristics

Morphological characteristics of the isolates like colony characters, Gram staining, Negative staining, Cell morphology, Cell motility, were observed (Krieg et al., 1984). Biochemical characteristics of the isolated strain were examined by Starch hydrolysis, Citrate utilization, Gelatin liquefaction, KOH test and Catalase test.

Inoculum preparation and xanthan gum production

A loopful of 72 hr old inoculum was transferred to 50 ml of YM broth (pH 7.0) and incubated at 37°C for 48 hr. About one ml of the *Xanthomonas sp*. culture was then transferred to 49ml of production medium (NH_4NO_3 -6 g/L, KH_2PO_4- 4 g/L,$MgSO_4$ -0.2 g/L, NaCl- 5 g/L, pH 7) supplemented with 10% of various carbon sources like molasses, sucrose , lactose and maltose in 100ml Erlenmeyer flask. The cultures were kept for incubation at 37°C/96 hr.

Biomass estimation

The biomass was calculated by measuring the dry weight of cell. About 5ml broth was separated by centrifuging at 10,000 rpm for 15 minutes. After centrifugation, supernatant containing xanthan gum was isolated from the pelleted biomass. Then the biomass was resuspended in deionized water and recentrifuged to precipitate the biomass. The stored biomass was dried in the hot air oven at 60°C for two hours and weighed.

Extraction of xanthan gum

The resulting supertant separated after biomass separation was added with 2 to 3 volumes of ethanol with continuous shaking so as to separate xanthan gum. Then obtained precipitate was separated by centrifugation at 6000 *rpm* for 15 min. The obtained deposit was transferred to a preweighed centrifuge tube and were kept in hot air oven for drying at 60°C for 20 hr.

Scanning electron microscopy

The structure of xanthan gum powder was characterized by estimations at 15 kV in an scanning electron microscope JSM 5800.

Analytical method

Investigation of functional group of synthesized xanthan gum was performed by FT-IR analysis. The dry powder of xanthan gum was incorporated with KBr and pressed into pellet under pressure and measured between the wavelength range of 4000 to 400 cm^{-1}.

Results and discussion

According to Gandhi et al. (1997) the carbon source impacts for cell development and the generation of the biopolymer; eventhough, given carbon source can support a efficient growth without huge production of polysaccharide. The fermentation of *X. campestris* by using sucrose as carbon source is the most suited for abundant xanthan production (Letisse et al., 2016; Souw and Demain., 1979). In the present work the biomass and xanthan gum production was examined by using various carbon sources like molasses, sucrose, lactose and maltose. The rate of xanthan gum generation during the cell development utilizing 10 g/L of different carbon sources was analysed. Molasses was found to be the most reasonable carbon source for biomass and xanthan gum generation by generating 5.6 g/L of biomass (Fig.1) and 3.11g/L of xanthan gum (Fig. 2). Minimal measure of xanthan gum production was found in the media supplemented with lactose (2.02g/l). This can happen because of the lack of migration of galactosidase synthesized by *X. campestris* which takes part in breaking lactose into glucose and galactose (Fu and Tseng., 1990). According to the perception of Krishna Leela and Gita Sharma., (2000) the biosynthesis of xanthan gum was maximum at 96 hr of incubation and there was no more improvement in viscosity of the polymer on prolonged incubation.

Whence observed on microscope, the cells of the isolate were Gram negative rod shaped and encircled by the slimy capsular layer. Slimy, yellow colored colonies were seen after plating on the 5% glucose supplemented medium. The biochemical test attributes revealed the isolate as *Xanthomonas sp*. (Table 1).

From the secured SEM image of xanthan gum it is visible that the gum is polygonal in shape that measures about 1 µm. (Fig. 3)

The Fourier Transform-infrared spectrum is an approach to find the resemblances or contrast in the chemical structure of the compounds. The uniqueness of the functional groups respective of the spectral bands of the synthesized xanthan gum within the range between 400 and 4000 cm^{-1} was correlated with data's obtained from Sandra et al., (2005) using commercial Xanthan. The most essential groups were in the mid region of 4000–400cm^{-1} and 3200–3450 cm^{-1} ;that denotes the axial distortion of –OH; 2850–2950 cm^{-1} that affirms axial deformation of C–H (might be because of assimilation of absorption of symmetrical and of asymmetrical stretching of CH_3 or even groups of CH_2) and CHO; 1710–1730 cm^{-1}represents axial twisting of C=O ester, acid, carboxylic, aldehydes and ketones; 1530–1650 cm^{-1} affirms axial disfigurement of C= O of enols (β-diketones) and 1050–1150 cm^{-1}predicts axial deformity of C–O.

The present research shows that the isolated polysaccharide followed the similar spectral behavior as the commercial. As per the perception of Faria et al., (2011) by Comparing the FTIR range of other commercially available gums, guar and xanthan, the spectrum finished up the bands around 3400 cm^{-1},2939 cm^{-1} and 990–1200 cm^{-1} that are normal to all polysaccharides predicts O–H bonds, C–H bonds of CH groups and saccharides, separately. (Fig. 3-7; Table-2)

Conclusion

Xanthan gum is a prevalent part of bacterial ooze. Efficient translation of carbon sources to the polysaccharide production requires a high carbon to nitrogen. The impact of various carbon hotspots for xanthan gum production utilizing *Xanthomonas* confines was resolved.

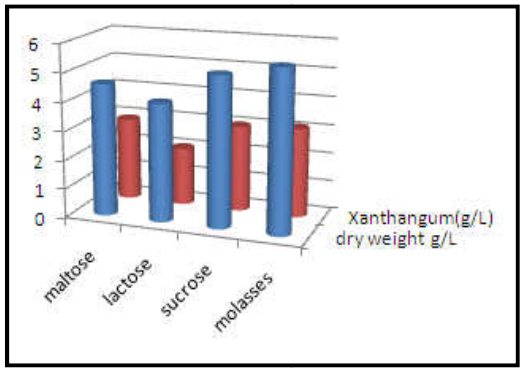

Fig. 1: Rate of xanthan gum produced from various carbon sources

Fig. 2: Xanthan gum after extraction with ethanol

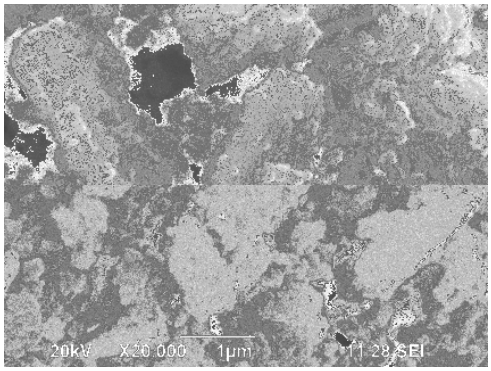

Fig. 3a: Scanning electron microscopic images of the xanthan gum produced

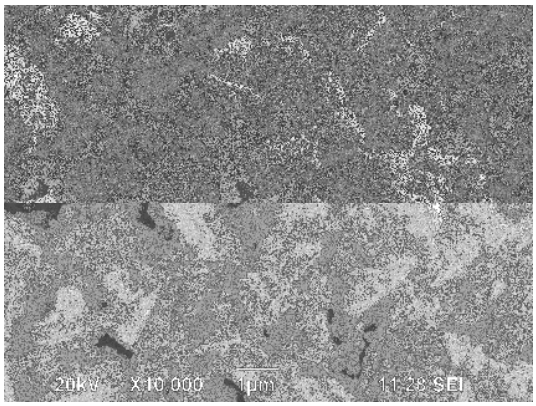

Fig. 3b: Scanning electron microscopic images of the xanthan gum produced

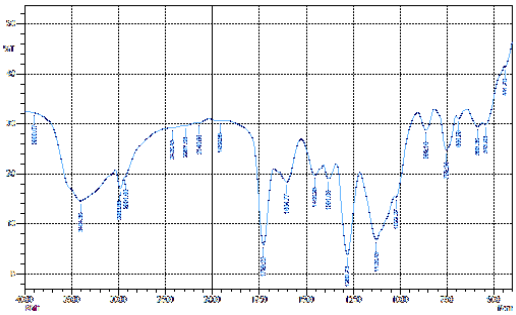

Fig. 4: FT-IR spectrum of xanthan gum produced by using molasses as carbon source

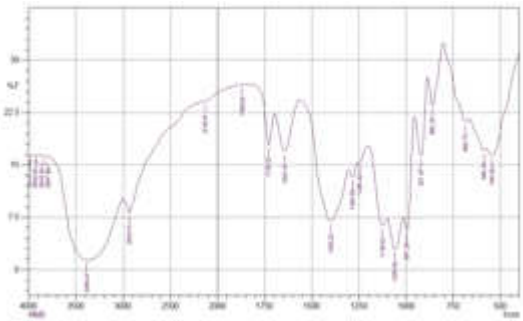

Fig. 5: FT-IR spectrum of xanthan gum produced by using Sucrose as carbon source

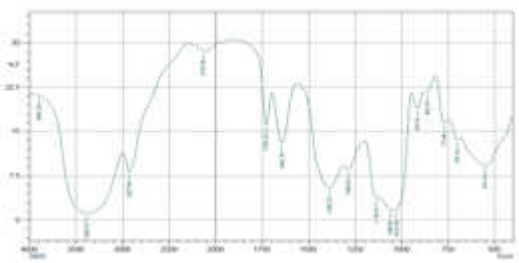

Fig. 6: FT-IR spectrum of xanthan gum produced by using Lactose as carbon source

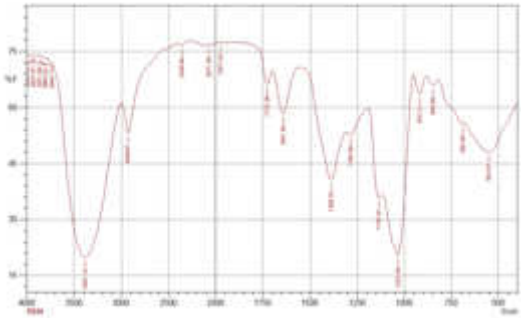

Fig. 7: FT-IR spectrum of xanthan gum produced by using Maltose as carbon source

Table 1: Biochemical characterization of *Xanthomonas sp*

S. No	Biochemical Test	Observation	Remarks
1.	Gram reaction	Gram Negative rod shaped cells observed	Gram Negative, Rods.
2.	Negative staining	Clear transparent zone around the cells was observed	Positive
3.	Starch Hydrolysis	Clear zone around the colonies on exposing to iodine vapours was observed.	Positive
4.	Citrate utilisation	The slant colour changed to deep Prussian blue after incubation.	Positive
5.	Kovacs' Oxidase	The appearance of cherry red ring on the surface of the incubated tube was observed.	Positive
6.	Gelatin Liquefaction	The organism found to liquefy gelatin.	Positive
7.	Fluorescent Pigmentation	No fluorescence was observed when placed under UV transillumination	Negative
9.	KOH test	Thread like slime was developed	Positive
10.	Catalase test	The isolate produced bubbles on exposure to Hydrogen peroxide.	Positive

Table 2: FT-IR interpretation of the xanthan gum produced by using different carbon sources

Functional Group	Transmittance Peak Observed				
	Commercial xanthan (cm^{-1})	RM1 (cm^{-1})	RM2 (cm^{-1})	RM3 (cm^{-1})	RM4 (cm^{-1})
Axial deformation of -OH	3200-3450	3404.26	3385.07	3385.07	3383.14
Axial deformation of C-H	2850–2950	2883.88 & 2931.8	2933.73	2927.94	2929.87
Axial deformation of C=O ester, acid carboxylic, aldehydes and ketones.	1710-1730	1728.22	1728.22	1728.22	1726.29
Axial deformation of C=O of enols	1530-1650	1604.77	1643.35	1643.35	1641.42
Axial deformation of C-H	1050-1150	1126.43	1058.92 & 1126.43	1058.92 & 1136.07	1136.07
Common to all polysaccharides	3400	3404.36	3385.07	3385.07	3383.14
Common to all polysaccharides	2939	2931.8	2933.73	2927.94	2929.87
Common to all polysaccharides	990-1200	1022.27& 1126.43,	997.2, 1058.92, 1126.43	1033.85, 1058.92 & 1136.07	1033.85 & 1136.07

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